Does LayerNorm in the critic reduce overestimation?

DDPG and TD3 on `LunarLander-v3`

Matteo Nizard
matteo.nizard@etu.sorbonne-universite.fr

Norman El Fartass
norman.el_fartass@etu.sorbonne-universite.fr

Abstract. Layer Normalization (LN) in the critic has been reported to reduce overestimation in SAC-based agents. We test this claim on DDPG and TD3 on `LunarLander-v3` (continuous), with 10 seeds per condition and a Monte Carlo measurement of the bias $Q - G$. TD3 already removes the bias, and LN changes neither its bias nor its return. When the bias comes back in TD3 without clipped double Q-learning, LN halves it ($+67 \to +36$), but in DDPG it increases it ($+114 \to +180$) and lowers the return. In our setting, the effect of LN on the bias therefore depends on the algorithm it is added to.

1 Introduction

In actor-critic methods the actor is designed to maximize the learned critic, so positive errors of Q are selected and then propagated by bootstrapping: the critic tends to *overestimate* the return of its own policy [3]. TD3 [3] was designed against this bias. More recently, a simpler solution has been reported: Layer Normalization (LN), introduced for supervised learning [1], placed in the critic. RLPD [2] argues that LN bounds the critic's values, which mitigates catastrophic overestimation on out-of-distribution actions. Nauman et al. [6] find, across 60+ SAC-based agents on DeepMind Control and Meta-World, that LN is among the most effective regularizers and outperforms methods designed against overestimation. However, these studies use SAC-based agents, often at high update-to-data (UTD) ratios, and none evaluates DDPG or TD3. We test whether this claim transfers to `LunarLander-v3` (continuous) at a UTD ratio of 1, and ask: what is the impact of LN in the critic on the overestimation bias of DDPG and TD3, and on their return?

2 Method

DDPG. We use DDPG [5]: a deterministic actor π_θ and a critic Q_ϕ with a target copy $Q_{\bar{\phi}}$. Unlike the original paper, the next action of the target is given by the *online* actor:

$$y = r + \gamma(1-d)\, Q_{\bar{\phi}}\big(s', \pi_\theta(s')\big), \tag{1}$$

$$\mathcal{L}(\phi) = \mathbb{E}_{\mathcal{B}}\big[(Q_\phi(s,a) - y)^2\big], \tag{2}$$

$$\nabla_\theta J = \mathbb{E}_{\mathcal{B}}\big[\nabla_a Q_\phi(s,a)\big|_{a=\pi_\theta(s)} \nabla_\theta \pi_\theta(s)\big], \tag{3}$$

with d the termination flag, soft target updates $\bar{\phi} \leftarrow \tau\phi + (1-\tau)\bar{\phi}$ and Gaussian exploration noise $\mathcal{N}(0, \sigma^2)$ (the environment clips actions to $[-1, 1]$).

TD3. Our implementation of TD3 [3] differs from this DDPG by four changes, each ablated in Section 5.1: (i) a target actor $\pi_{\bar{\theta}}$; (ii) target policy smoothing, $\tilde{a}' = \pi_{\bar{\theta}}(s') + \mathrm{clip}(\varepsilon, -c, c)$, $\varepsilon \sim \mathcal{N}(0, \tilde{\sigma}^2)$; (iii) double Q-learning: both critics regress on $y = r + \gamma(1-d)\min_{i=1,2} Q_{\bar{\phi}_i}(s', \tilde{a}')$; (iv) delayed policy updates: the actor (which maximizes Q_{ϕ_1}) and the targets are updated every $d_\pi = 2$ critic updates.

LayerNorm in the critic. For a pre-activation $z \in \mathbb{R}^n$, $\mathrm{LN}(z) = g \odot (z - \mu(z))/\sqrt{\sigma^2(z) + \epsilon} + b$, with μ, σ^2 the mean and variance over the n units and a learnable gain g and bias b. We compare a critic without LN to three placements of LN in the critic (in both critics for TD3): before the ReLU of every hidden layer, after the ReLU of every hidden layer, and on the last hidden layer only, before its ReLU (Figure 1). LN is never applied to the output of the critic, and the actor is left unchanged.

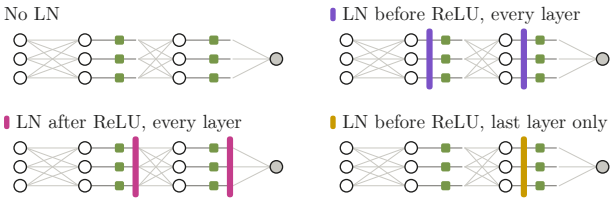


Figure 1: The critic without LN and with the three LN placements (illustrative: (s, a) on the left, Q on the right, 3 units per hidden layer). Green squares: ReLU, on each unit; colored bars: LN, on the whole layer.

In all three LN placements h_2 is a normalized vector, or its ReLU, so $\|h_2\| \le \sqrt{n}\,\|g_2\|_\infty + \|b_2\|$ and $|Q(s,a)| \le \|w_3\|\big(\sqrt{n}\,\|g_2\|_\infty + \|b_2\|\big) + |b_3|$ for every (s,a): this is the bound behind the RLPD argument. Without LN, no such bound exists.

Section 5.2 compares the three placements to the baseline without LN. Everywhere else, "LN" means the first placement (before the ReLU of every hidden layer, the convention of [6]), except in Figure 2, which pools the three placements.

Measuring the bias. We compare the critic to the true return of the current policy, estimated by Monte Carlo (using the policy learned). At each evaluation pause (Section 3), the deterministic policy plays 10 bias episodes from fixed start states. For each state s_t visited in the bias episodes, we compare the critic's estimate to the discounted return actually obtained afterwards:

$$\text{bias} = \frac{1}{N}\sum_t \big(Q_{\phi_1}(s_t, \pi_\theta(s_t)) - G_t\big), \quad G_t = \sum_{k \ge 0} \gamma^k r_{t+k}, \tag{4}$$

where N is the number of states kept over the 10 episodes. Since `LunarLander-v3` (without wind) and the policy are deterministic, a single rollout gives the exact return of each state-action pair. A positive bias means that the critic overestimates the return of its own policy. For TD3 we only measure Q_1.

Two details make this measurement reliable. First, episodes are truncated at 1000 steps if they do not finish. For a state close to the cutoff, G_t misses the rewards the agent would still collect after the limit, which artificially inflates $Q - G$. We therefore drop the last H steps of truncated episodes, with H chosen such that $\gamma^H \leq 0.05$ (the missing rewards then weigh at most 5%); episodes that terminate naturally are kept entirely, since no reward follows their end. Second, we report the absolute bias $Q-G$ rather than the normalized bias $(Q-G)/|G|$, which is unstable when returns are close to 0.

Statistical protocol. Each condition is run with seeds 1–10. We only plot the 39 measurements (see Section 3) we gathered from the training pauses as a curve (Figure 2): they show the dynamics but are not statistical tests. For the tests, we use the final return of the evaluations at training step 190k, 195k and 200k as means (30 episodes in total) and the final bias at steps 180k to 200k as the mean of bias test, which smooths the noise between consecutive measurements. These values are compared with Welch's two-sided t-test [8], which does not assume equal variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}, \qquad \nu \approx \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}. \quad (5)$$

We report the raw p-value of each test and call a difference significant when $p < 0.05$, without correction for multiple comparisons.

3 Experimental setup

`LunarLander-v3` with `continuous=True` (observation dimension 8, two actions in $[-1, 1]$), 200k environment steps per run, 10k warm-up steps, UTD ratio 1. Every 5k environment steps from 10k to 200k (39 times per run), training is paused. During a pause, the deterministic policy (no exploration noise) plays, in separate environments, for 10 evaluation episodes (return only) and 10 bias episodes. The evaluation episodes start from new states at each pause, while the bias episodes always start from the same 10 states. All start states depend only on the seed, so for a given seed every condition is evaluated from exactly the same start states. Both TD3 and DDPG algorithms use the same hyperparameters (Appendix B): two hidden layers of 400 and 300 units, learning rates 10^{-3}, $\gamma = 0.98$, batch 256, $\tau = 0.005$, $\sigma = 0.1$, taken from RL Baselines3 Zoo [7]. This configuration was selected among three candidates (the other two in Table 4) for its return without LN in a 3-seed pilot. The main experiment crosses {DDPG, TD3} × {no LN, LN}; a second campaign replicates it and adds the TD3 ablations, LN after the ReLU and the four conditions under the two other configurations, and a third one tests LN on the last hidden layer only. All runs use CPUs on shared university lab machines that are only available at night and on weekends. One run takes about 40 minutes and the 348 runs total about 230 CPU-hours. This budget set the design we used. Code, campaign specifications and raw results: `https://github.com/matleniz/ddpg-td3-layernorm-bias`.

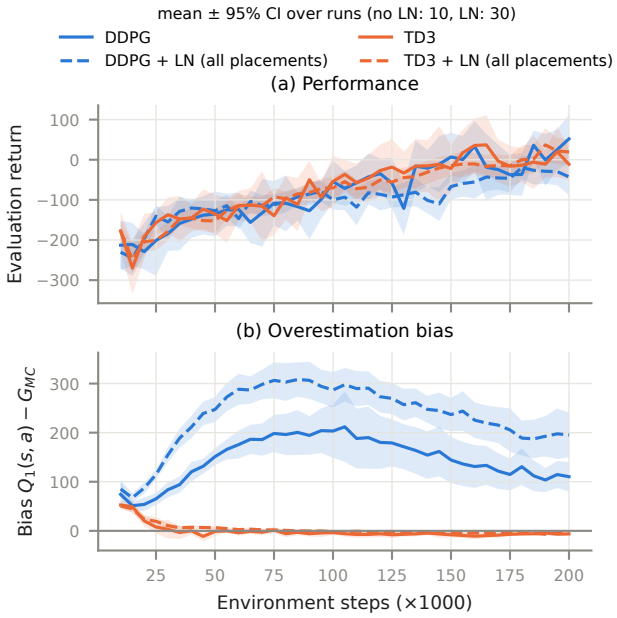


Figure 2: (a) Evaluation return and (b) bias $Q_1(s, a) - G$, measured at each training pause (every 5k steps). We plot the mean of the 10 evaluation episodes played during the pause. Curves: mean ± 95% CI over runs (10 without LN; 30 with LN, pooling the three placements). LN raises the bias of DDPG and lowers its return, and leaves TD3 unchanged (Welch tests on the final values: Table 1).

Condition	Final return	Final bias $Q_1 - G$
DDPG	$+25.5 \pm 63.4$	$+114.3 \pm 37.8$
DDPG + LN	-63.6 ± 56.8	$+179.5 \pm 40.4$
TD3	-0.8 ± 45.0	-6.0 ± 3.1
TD3 + LN	$+16.2 \pm 69.2$	-5.7 ± 2.8
Welch test	p (return)	p (bias)
DDPG vs DDPG + LN	**0.004**	**0.002**
TD3 vs TD3 + LN	0.525	0.831
DDPG vs TD3	0.300	$< 10^{-4}$

Table 1: Final return and final bias (mean ± std over 10 seeds) and Welch tests. Bold: $p < 0.05$.

4 Results

Effect of LN. Figure 2 shows the dynamics, with LN pooled over the three placements. The tests of Table 1 use the main placement only (before ReLU, every layer): with this LN, the bias of DDPG increases from +114 to +180 ($p = 0.002$) and its final return drops from +25.5 to −63.6 ($p = 0.004$). For TD3, LN changes neither the bias (−6.0 vs −5.7, $p = 0.83$) nor the return ($p = 0.52$). TD3 removes the overestimation of DDPG (bias −6 vs +114, $p < 10^{-4}$), but its final return is not significantly different ($p = 0.30$). Under the two other configurations (Table 4), LN again has no significant effect on the bias of TD3; for DDPG it increases the bias with `fujimoto` (+251 → +379, $p = 0.019$) but slightly decreases it with `tp04` (+34 → +26, $p = 0.034$).

5 Analysis

LN reduces the bias of no agent of Table 1. To find a case where it can, we first look for the TD3 component that removes the bias (Section 5.1). This gives a TD3 variant that overestimates like DDPG, in which LN helps.

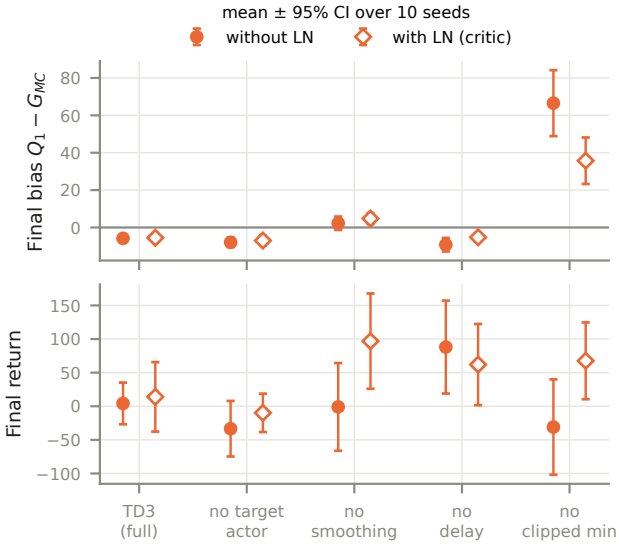


Figure 3: TD3 with one component removed, without (filled) and with (hollow) LN in the critic. Final bias (top) and final return (bottom), mean ± 95% CI over 10 seeds (final: last 5 bias measurements, last 3 evaluations). Only removing the min brings the bias back; there, LN halves it ($p = 0.005$) and raises the return ($p = 0.025$). Welch tests: Table 2.

Agent	LN	Bias	p	Return	p
DDPG	all	+114→+184	**0.001**	+24→−58	**0.008**
TD3	all	−6→−6	0.818	+4→+14	0.719
TD3 −tgt. actor	all	−8→−7	0.561	−33→−10	0.304
TD3 −smoothing	all	+2→+5	0.236	−1→+97	**0.034**
TD3 −delay	all	−9→−5	**0.046**	+88→+62	0.529
TD3 −min	all	+67→+36	**0.005**	−31→+68	**0.025**
DDPG	after	+114→+144	0.120	+24→−1	0.420
TD3	after	−6→−5	0.399	+4→+31	0.407
DDPG	last	+114→+253	**0.030**	+24→−36	0.095
TD3	last	−6→−5	0.686	+4→+31	0.459
TD3 −min	last	+67→+30	**0.001**	−31→+62	**0.039**

Table 2: Effect of LN in the critic, second and third campaigns: final bias and return without → with LN (mean over 10 seeds). LN in every hidden layer before (all) or after (after) the ReLU, or in the last hidden layer only (last); TD3 $-x$: TD3 without component x. p: Welch test. Bold: $p < 0.05$.

Section 5.2 checks that the conclusion does not depend on where LN is placed in the critic, and Section 5.3 lists the placements we did not test.

5.1 TD3 ablations: which component removes the bias?

Removing double Q-learning is the only ablation that brings back overestimation (bias +67, Figure 3); with any other component removed the bias stays within 10 points of 0, and LN moves it by at most 4 points. Without the min, LN halves the bias (+67 → +36, $p = 0.005$, Table 2), in line with Nauman et al. [6], and the return increases (−31 → +68, $p = 0.025$). Without the min, the two critics no longer interact: Q_{ϕ_1} regresses on its own target and is the only critic the actor uses. TD3 −min is therefore DDPG plus a target actor, target smoothing and delayed updates. Both overestimate without LN, yet LN reduces the bias of TD3 −min and increases that of DDPG: one or several of these three components changes the effect of LN. Our runs do not tell which one, and we do not attempt to explain the difference; this would require DDPG with one component added at a time, with and without LN.

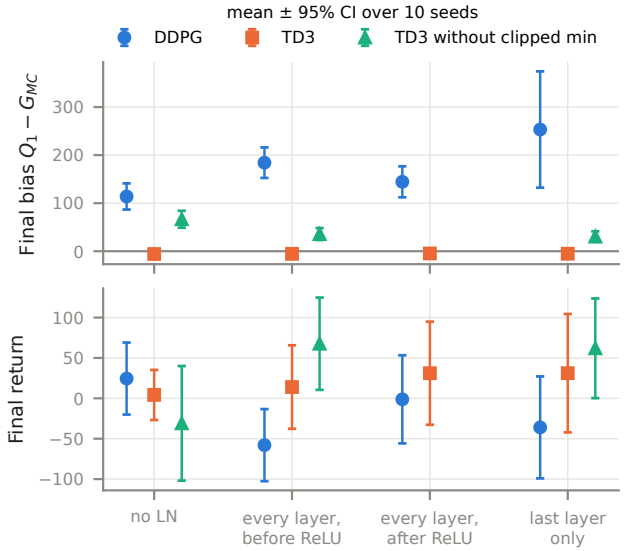


Figure 4: LN placement: none, before or after the ReLU of every hidden layer, or before the ReLU of the last hidden layer only. Final bias (top) and final return (bottom), mean ± 95% CI over 10 seeds. No placement reduces the bias of DDPG or TD3; both placements tested on TD3 −min reduce it ($p = 0.005, 0.001$). Welch tests: Table 2.

5.2 Where to put the LN

The placement does not change the conclusion (Figure 4, Table 2). A single LN on the last hidden layer, the only one the RLPD bound and the PQN result rely on, reduces the bias of TD3 −min as much as LN in every layer (+67 → +30, $p = 0.001$), and has no effect on TD3 (−5.8 → −5.2). For DDPG it gives the highest bias of all placements (+114 → +253, $p = 0.030$), with a large spread across seeds (std 169 vs 38). LN after the ReLU increases the DDPG bias less (+114 → +144, $p = 0.12$). So a single LN in front of the output, the one that bounds Q, is enough to reduce the bias of TD3 −min, and it does not prevent the increase in DDPG.

5.3 Placements we did not test

All our runs put LN in the critic only. LN could also be placed elsewhere, first in the actor: in the hidden layers of the actor alone, or in both the actor and the critic. Our code supports it, but we did not run it within the budget, so we cannot say whether LN in the actor changes the bias. Other candidates are LN on the input of the critic (the state-action pair) or in the target networks only.

6 Conclusion and limitations

In our setting, LN in the critic is not a general remedy for overestimation. With double Q-learning, TD3 has almost no bias and LN has no detectable effect. Without the min, LN halves the bias, as reported for SAC [6]; in DDPG it increases the bias and lowers the return. Whether LN reduces the bias thus depends on the algorithm it is added to, and our runs do not identify which component of TD3 makes the difference.

Most limitations come from the compute budget (about 230 CPU-hours on machines available only at night and on weekends, over a few days), which ruled out longer

runs, more seeds and more environments. (i) At 200k steps no condition reaches the "solved" threshold of +200; the effect of LN may differ later in training. (ii) With a standard deviation of the final return of 40–100 across seeds, 10 seeds only detect, with 80% power, differences larger than about 1.3 standard deviations (60 to 130 points); the absence of an effect of LN on TD3 is a lack of evidence, not evidence of absence. (iii) We run 36 tests without correction for multiple comparisons: at the 5% level, about two false positives are expected, so isolated p-values just below 0.05 (e.g. the return gains of TD3 −smoothing and TD3 −min with LN) should be read with caution. (iv) We study a single environment and a UTD ratio of 1, whereas the LN literature mostly uses high UTD ratios. (v) LN is only placed in the critic (Section 5.3).

References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.

[2] Philip J. Ball, Laura Smith, Ilya Kostrikov, and Sergey Levine. Efficient online reinforcement learning with offline data. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2302.02948.

[3] Scott Fujimoto, Herke van Hoof, and David Meger. Addressing function approximation error in actor-critic methods. In *International Conference on Machine Learning (ICML)*, 2018. arXiv:1802.09477.

[4] Matteo Gallici, Mattie Fellows, Benjamin Ellis, Bartomeu Pou, Ivan Masmitja, Jakob Nicolaus Foerster, and Mario Martin. Simplifying deep temporal difference learning. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2407.04811.

[5] Timothy P. Lillicrap, Jonathan J. Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2016. arXiv:1509.02971.

[6] Michal Nauman, Michał Bortkiewicz, Piotr Miłoś, Tomasz Trzciński, Mateusz Ostaszewski, and Marek Cygan. Overestimation, overfitting, and plasticity in actor-critic: the bitter lesson of reinforcement learning. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2403.00514.

[7] Antonin Raffin. RL baselines3 zoo. `https://github.com/DLR-RM/rl-baselines3-zoo`, 2020.

[8] Bernard L. Welch. The generalization of 'Student's' problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947.

A Claims of the literature on LN in the critic

RLPD [2]. Online RL with offline data. With normalized features, $\|Q(s,a)\| \leq \|w\|$ for any action, including out-of-distribution ones, which "mitigates catastrophic overestimation". Crucial on sparse-reward Adroit with little data, helps on AntMaze; on locomotion its impact is "not so clear cut".

Nauman et al. [6]. Large study of 60+ SAC-based agents; LN applied to each hidden layer is among the most effective regularizers and outperforms methods designed against overestimation, including clipped double Q-learning. Strong on DMC Dog, poor on MetaWorld.

PQN [4]. LN makes TD learning provably stable for wide networks, and only the final layer weights matter for stability; an ℓ_2 penalty on these weights is recommended only when LN alone does not stabilize training (in their experiments, only on Baird's counterexample). Stability does not imply the absence of a maximization bias.

B Hyperparameters

Parameter	Value (DDPG and TD3)
Environment	`LunarLander-v3`, `continuous=True`
Env. steps / warm-up	200k / 10k
Replay buffer	200k (whole run)
γ / batch / τ	0.98 / 256 / 0.005
Actor, critic hidden layers	400–300
Actor, critic learning rate	10^{-3}
Exploration noise σ / UTD	0.1 / 1
TD3: d_π / $\tilde{\sigma}$ / c	2 / 0.2 / 0.5
LayerNorm	critic(s), before ReLU, all hidden
Evaluation	every 5k steps, 10 episodes
Seeds	1–10

Table 3: Hyperparameters (all conditions unless stated otherwise).

C Additional results

Agent	LN	Bias	p	Return	p
DDPG (fujimoto)	all	+251→+379	**0.019**	−72→−92	0.560
TD3 (fujimoto)	all	−11→−7	0.445	−30→+16	0.291
DDPG (TP04 of RL course)	all	+34→+26	**0.034**	+11→0	0.752
TD3 (TP04 of RL course)	all	−4→−2	0.165	+35→−57	**0.015**

Table 4: Effect of LN under two other configurations: `fujimoto` [3] and `TP04` of RL course (2 HL of 64 neurons each). 10 seeds each, Welch tests.